

Suwisa Aitha

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EDUCATION

George Mason University
M.S. in Electrical Engineering

Fairfax, VA
Jan 2025 – Graduated May 2026

Khon Kaen University
B.S. in Electrical Engineering

Khon Kaen, Thailand
Aug 2018 – Jul 2022

TECHNICAL SKILLS

Electrical Engineering: Power Flow Analysis, Fault Analysis, Power System Protection, Stability Analysis, Frequency Response, RoCoF Analysis, Harmonic Analysis (THD)

Software: MATLAB/Simulink, PowerWorld, AutoCAD, Python, MS Office Suite

Hardware: ESP32, Arduino

PROFESSIONAL EXPERIENCE

Electrical Design and Visualization (Co-op)
B.S.E. Electronics Co., Ltd.

Bangkok, Thailand
Jul 2021 – Nov 2021

- Developed electrical layouts and visualization models using AutoCAD to support electrical system integration
- Compared simulation results with physical measurements and achieved approximately 87% correlation for system validation
- Assisted in preparing technical documentation, electrical configurations, and engineering support tasks
- Supported coordination between design and implementation processes for electrical systems

ENGINEERING PROJECT EXPERIENCE

Grid-Forming Inverter with Virtual Synchronous Machine (VSM) Control **George Mason University**

- Developed a grid-forming inverter model with VSM control in MATLAB/Simulink for inverter-based microgrids
- Simulated virtual inertia and frequency support to improve renewable energy integration and grid resilience
- Analyzed inverter synchronization, transient response, and frequency regulation under dynamic conditions

Field Oriented Control (FOC) of Three-Phase Induction Motor – MATLAB/Simulink **George Mason University**

- Designed and implemented FOC for a three-phase induction motor using MATLAB/Simulink
- Applied Clarke and Park transformations, PI controllers, and PWM inverter control for speed regulation
- Evaluated motor speed tracking, current response, and dynamic performance under varying conditions

Unified Power Quality Conditioner (UPQC) **George Mason University**

- Designed and simulated a UPQC system in MATLAB/Simulink for voltage sag and harmonic mitigation
- Reduced total harmonic distortion (THD) by approximately 53% after compensation
- Evaluated system performance under dynamic load conditions

Power System Dynamics with Renewable Energy Penetration - Simulation Project **George Mason University**

- Developed a Single Machine Infinite Bus (SMIB) model in MATLAB/Simulink for power system dynamic analysis
- Evaluated frequency nadir, rotor angle stability, settling time, and RoCoF during disturbances
- Assessed the impact of renewable energy penetration on power system stability and dynamic performance

CERTIFICATIONS & TRAINING

- Dominion Energy Fault Analysis Certificate Program – Completed
- FE (Fundamentals of Engineering) Exam – Scheduled July 22, 2026

AWARDS & RECOGNITION

- Mason Recreation Fitness Employee of the Year 2025 – 2026